Nonparametric Methods for a Single Unimodal Sample

(Fisher, *Statistical Analysis of Circular Data*, 1993, pp. 72–81)

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Set-up & notation

Data: i.i.d. angles $\{\theta_1, \ldots, \theta_n\} \in (-\pi, \pi]$ (or $[0, 2\pi)$).
Resultant components and length:

$$C = \sum_{i=1}^{n} \cos\theta_i, \quad S = \sum_{i=1}^{n} \sin\theta_i, \quad R = \sqrt{C^2 + S^2}, \quad \bar{R} = R/n.$$

Sample mean direction (equals the direction of the resultant):

$$\hat{\mu} \text{ with } \cos\hat{\mu} = C/R, \ \sin\hat{\mu} = S/R.$$

Sample circular dispersion (plays the role of variance in large-sample inference):

$$\hat{\delta} = \frac{1 - \hat{\rho}_2}{2\bar{R}^2}, \quad \hat{\rho}_2 = \sqrt{C_2^2 + S_2^2}, \quad C_2 = \frac{1}{n}\sum \cos(2\theta_i), \ S_2 = \frac{1}{n}\sum \sin(2\theta_i).$$

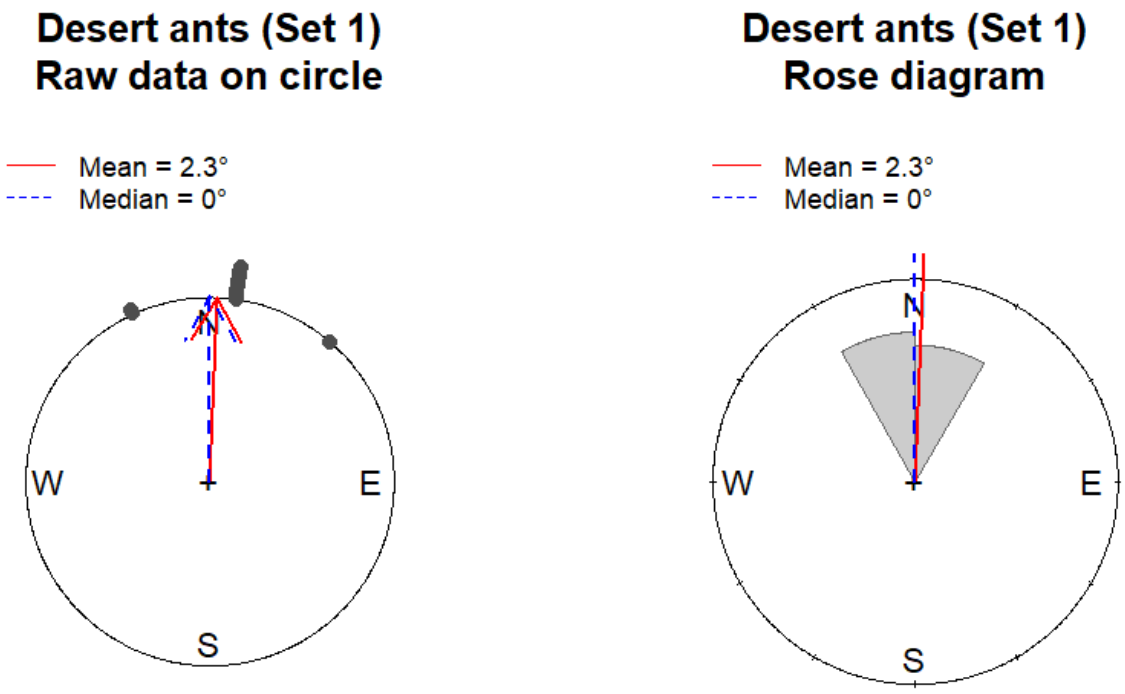
Population parameter of interest: the (unknown) mean/median direction.
Goal: inference on a "central" direction under unimodality without fully specifying a parametric model.

Why nonparametrics here?

- Under a single-peaked (unimodal) pattern, we often want a robust "center".
- Two natural centers on the circle:
  - $\tilde{\theta}$: *sample median direction* (robust, exact small-sample CI available).
  - $\hat{\mu}$: *sample mean direction* (efficient when concentration is clear; large-sample normal approximations available).
- Strategy (Fisher Ch. 4.4): use median-based exact/sign-type methods for small n; use $\hat{\mu}$ with large-sample formulae when n is moderate/large or concentration is not too weak.

Desert ants (Set 1): mean and median

Figure: Raw circular plot (left) and rose diagram (right) for the desert ants data (Set 1). The red arrow shows the sample mean direction ($\approx 2.3°$) and the blue dashed arrow shows the sample median direction ($= 0°$).



Median direction: definition

Definition (population): there exists unique direction ϕ splitting probability mass in half (unimodal case).
Sample median direction $\tilde{\theta}$: choose $\tilde{\theta}$ so that at least half of the sample lies within any semicircle centered at $\tilde{\theta}$.
Intuition: like the linear median but computed on the circle; robust to outliers and skew.

CI for the median direction

Construction. For any integer $m = 1, 2, \ldots$, count off m θ-values to the left and to the right of the sample median direction $\tilde{\theta}$ (not counting $\tilde{\theta}$ itself) to obtain the lower and upper data values $\theta_{(L_m)}$ and $\theta_{(U_m)}$. The open arc $\big(\theta_{(L_m)}, \theta_{(U_m)}\big)$ containing $\tilde{\theta}$ is then a confidence interval for the population median direction ϕ, with confidence level $1 - \alpha$ determined by m.

(a) Small samples: $n < 16$**.** Appendix A6 lists admissible values of m together with the corresponding *exact* confidence levels for the interval $\big(\theta_{(L_m)}, \theta_{(U_m)}\big)$.

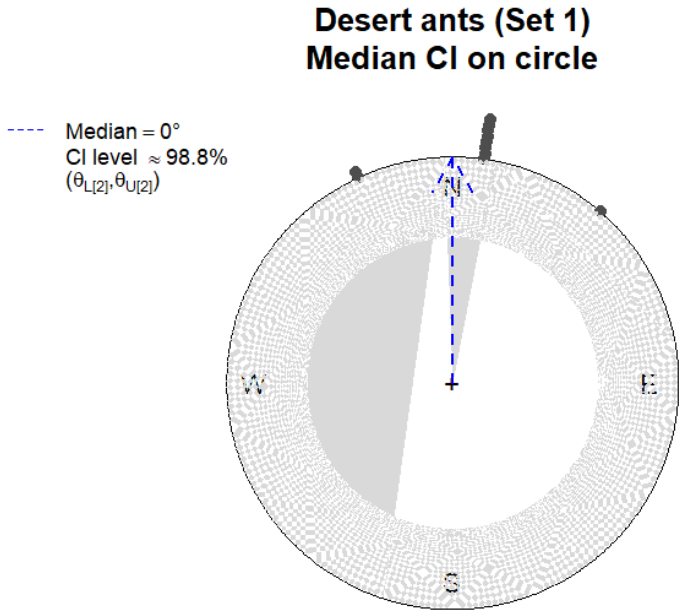
(b) Moderate/large samples: $n \geq 16$**.** An approximate $100(1-\alpha)\%$ confidence interval is obtained by setting

$$m \;=\; 1 + \left\lfloor \tfrac{1}{2}\, z_{\alpha/2}\, \sqrt{n} \right\rfloor, \tag{4.19}$$

where $z_{\alpha/2}$ is the upper $100(\alpha/2)\%$ point of the $N(0,1)$ distribution (Appendix A1).

Median direction: exact CI (Desert ants, Set 1)

Figure: Exact confidence interval for the median direction in the Desert ants data (Set 1). The dashed blue arrow shows the sample median direction $\tilde{\theta} = 0°$; the grey sector represents the $(\theta_{(L_{[2]})}, \theta_{(U_{[2]})})$ interval with confidence level $\approx 98.8\%$.



Testing a specified median direction

We test $H_0: \ \tilde{\mu} = \tilde{\mu_0}$ against $H_1: \ \tilde{\mu} \neq \tilde{\mu_0}$ at level α. Define, using indicators,

$$k \ = \ \sum_{i=1}^{n} \mathbf{1}\{(\theta_i - \tilde{\mu}_0) \bmod 2\pi = 0\}\,, \qquad m \ = \ \sum_{i=1}^{n} \mathbf{1}\{(\theta_i - \tilde{\mu_0}) \bmod 2\pi \in (0, \pi)\}\,.$$

Two procedures.

1. *CI inclusion method.($n < 16$)* If a $100(1-\alpha)\%$ CI for the median direction $\tilde{\mu}$ has been computed, **reject** H_0 when $\tilde{\mu_0}$ lies *outside* this CI.
2. *Direct large-sample test* (recommended when $n \geq 16$). Let

$$\boxed{Y = \frac{(2m - n + k)^2}{n - k}} \tag{4.20}$$

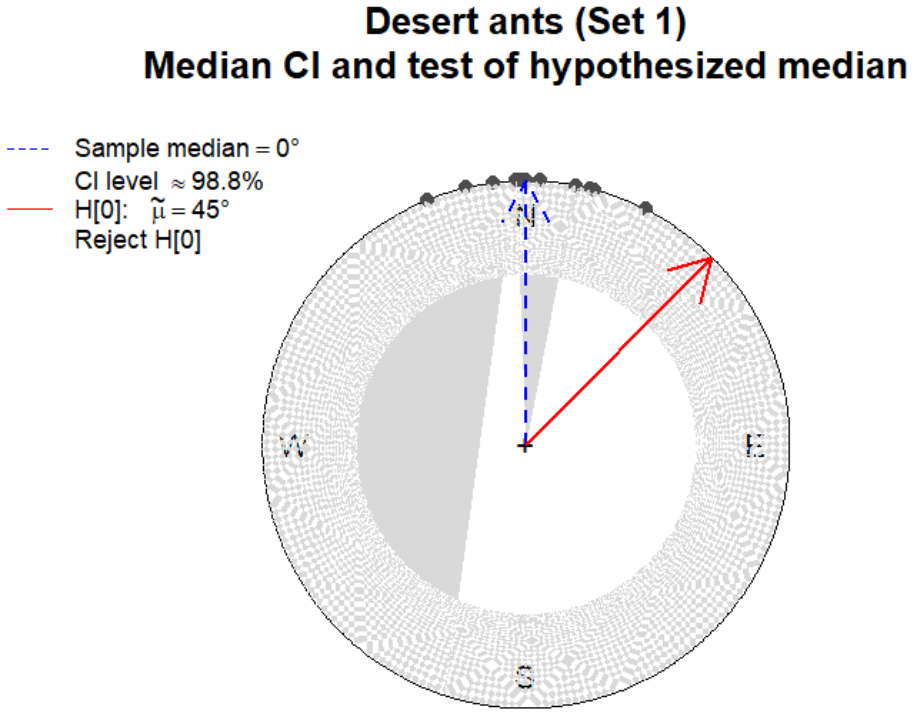
Under H_0, $Y \approx \chi^2_1$. **Reject** H_0 at level α if

$$Y \ > \ \chi^2_{1,\,\alpha} \quad \text{(Appendix A2).}$$

Notes. $n - k \geq 16$ at least.

Median direction CI and hypothesis test (Desert ants, Set 1)

Figure: Exact confidence interval for the sample median direction in the Desert ants data (Set 1). The dashed blue arrow shows the sample median $\tilde{\theta} = 0°$; the grey sector depicts the $(\theta_{(L_{[2]})}, \theta_{(U_{[2]})})$ interval (CI $\approx 98.8\%$). The red arrow shows the hypothesized median $\tilde{\mu}_0 = 45°$, which lies outside the CI, leading to rejection of H_0.



Mean direction: CI

Point estimate: $\hat{\mu}$ (direction of the resultant).
Two ways to quantify uncertainty:

1. **Small n($n < 25$): bootstrap CI**. Resample $\{\theta_1, \ldots, \theta_n\}$ with replacement, compute $\hat{\mu}^*$, and use the bootstrap percentiles for a CI of μ.
2. **Large n($n \geq 25$): analytic (normal) CI.** Compute the circular dispersion $\hat{\delta}$ as in (2.28), and the circular standard error of $\hat{\mu}$ via

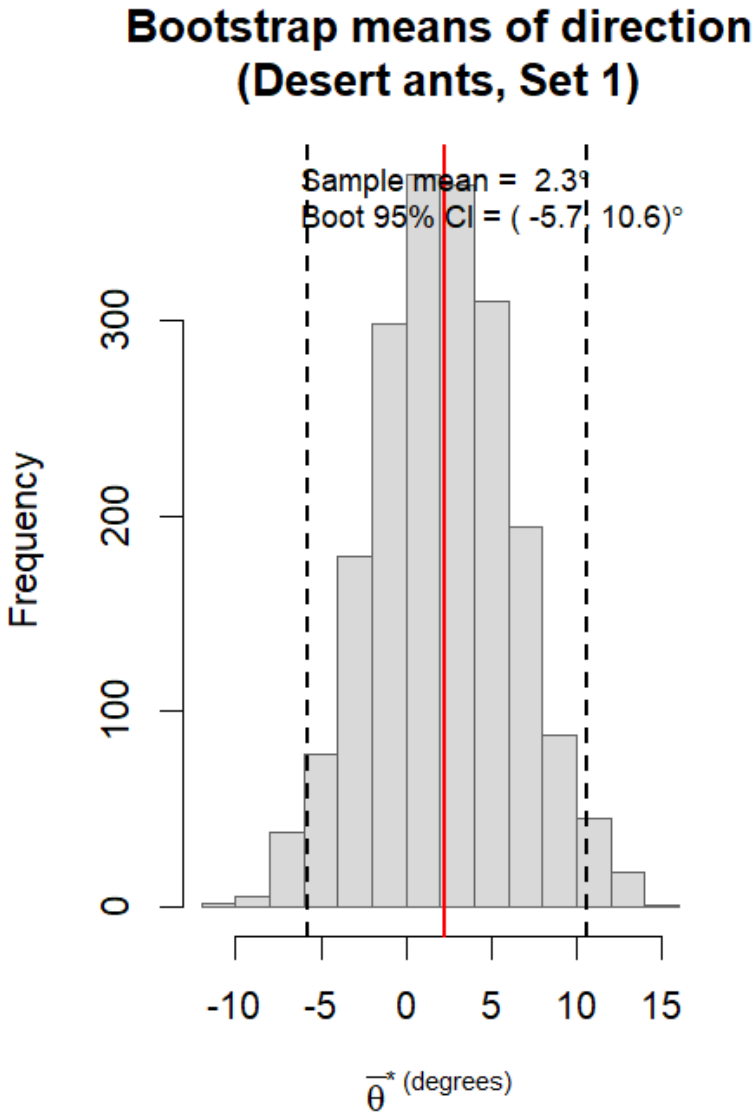
$$\hat{\sigma}^2 = \frac{\hat{\delta}}{n}. \tag{4.21}$$

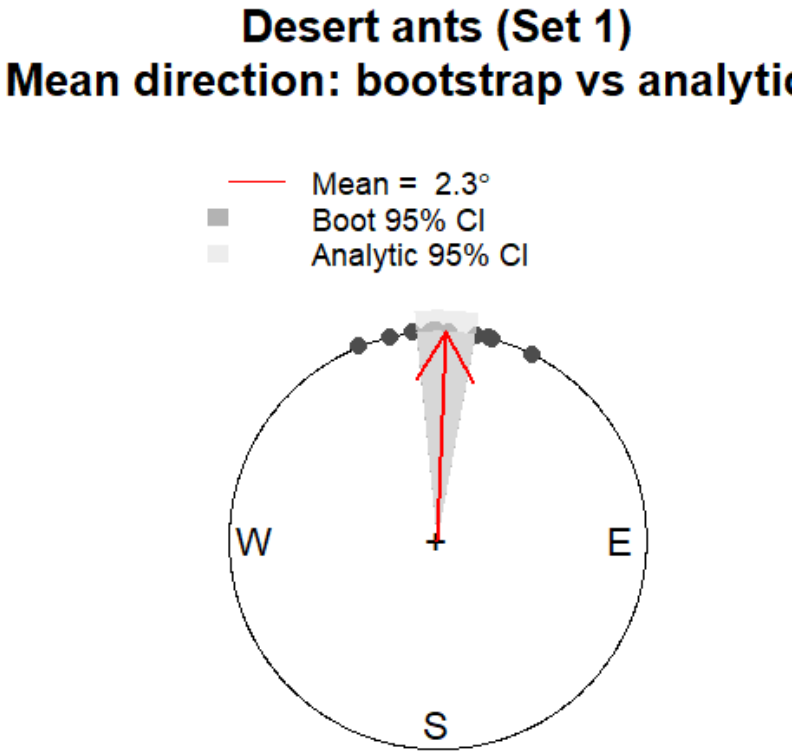
Then an approximate $100(1-\alpha)\%$ confidence interval for μ is

$$\Big(\hat{\mu} - \sin^{-1}\big(z_{\alpha/2}\,\hat{\sigma}\big), \;\; \hat{\mu} + \sin^{-1}\big(z_{\alpha/2}\,\hat{\sigma}\big) \Big), \tag{4.22}$$

where $\hat{\delta} = \frac{1-\hat{\rho}_2}{2\bar{R}^2}$.

Mean direction: bootstrap vs analytic CI (Desert ants, Set 1)



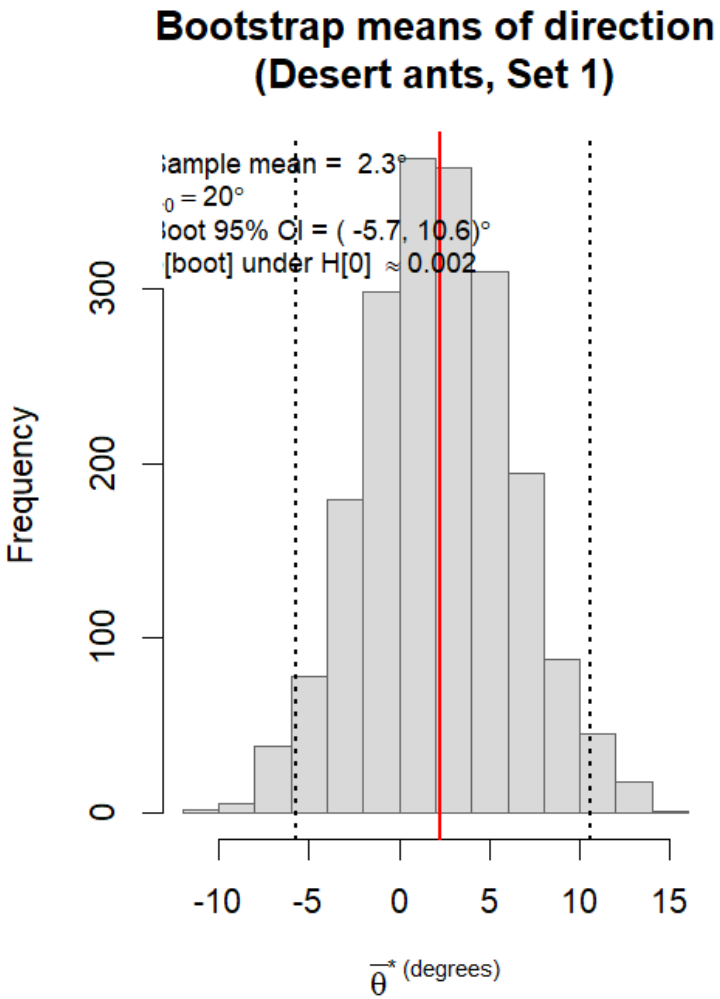


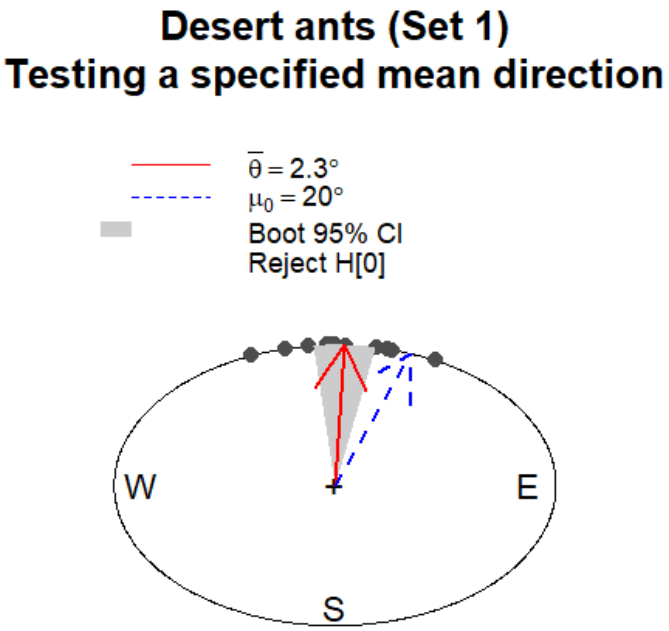
Testing a specified mean direction

We wish to test $H_0: \ \mu = \mu_0$ against $H_1: \ \mu \neq \mu_0$ at level α. There are two approaches:

1. **CI inclusion method (all n).** If a $100(1-\alpha)\%$ confidence interval for μ has already been computed (as in §4.4.4), **accept** H_0 at the $100(1-\alpha)\%$ level when μ_0 lies inside that interval; otherwise **reject** H_0.
2. **Direct test (depends on n).**
   (a) $n < 25$: use a *bootstrap* procedure (see §8.4). One may decide whether to assume symmetry of the underlying distribution about its mean direction using the methods in §4.4.6.
   (b) $n \geq 25$: compute the circular standard error a from (4.21), using $\hat{\delta}$ from (2.28):
   $$\hat{\sigma}^2 = \frac{\hat{\delta}}{n}. \tag{4.21}$$
   Then form the test statistic
   $$S \ = \ \frac{\sin(\hat{\mu} - \mu_0)}{\hat{\sigma}}. \tag{4.23}$$
   Critical values. Reject H_0 at level α if $|S| > z_{\alpha/2}$; here $z_{\alpha/2}$ is the upper $100(\alpha/2)\%$ point of the $N(0,1)$ distribution (Appendix A1).

Testing a specified mean direction (Desert ants, Set 1)





Testing symmetry about an unspecified direction

Test whether the underlying distribution is symmetric about *some* direction.
Construction (Wilcoxon signed-rank on circular deviations).

1. Compute the sample median direction $\tilde{\theta}$ (the "median vector").
2. Define wrapped signed deviations $\phi_i = \theta_i - \tilde{\theta}$. Delete values with $\phi_i = 0$; let n' be the remaining sample size.
3. Rank the absolute deviations: $r_i = \text{rank}\left(|\phi_i|\right)$ among the nonzero $|\phi|$'s, using average ranks for ties.
4. Form the Wilcoxon sum of positive ranks

$$w_{n'}^{+} = \sum_{\phi_i > 0} r_i. \tag{4.26}$$

Testing symmetry about an unspecified direction

Critical values.

(a) $n' < 16$: For a level-α test, **reject** symmetry if

$$w_{n'}^{+} \;\geq\; w_{n',\alpha/2} \quad \text{or} \quad w_{n'}^{+} \;\leq\; \frac{n'(n'+1)}{2} - w_{n',\alpha/2}, \tag{4.27}$$

where $w_{n',\alpha/2}$ is the tabulated $\alpha/2$ critical value (Appendix A7).

(b) $n' \geq 16$: Use the normal approximation

$$w^{+} \;=\; \frac{w_{n}^{+} - \frac{n'(n'+1)}{4} + \frac{1}{2}}{\sqrt{\frac{n'(n'+1)(2n'+1)}{24}}} \;,$$

and **reject** for large $|w^{+}|$; equivalently, report the two-sided $p = 2\,[1 - \Phi(w^{+})]$.

Notes. This test is distribution-free under symmetry.

Testing symmetry about an unspecified direction (Desert ants, Set 1)

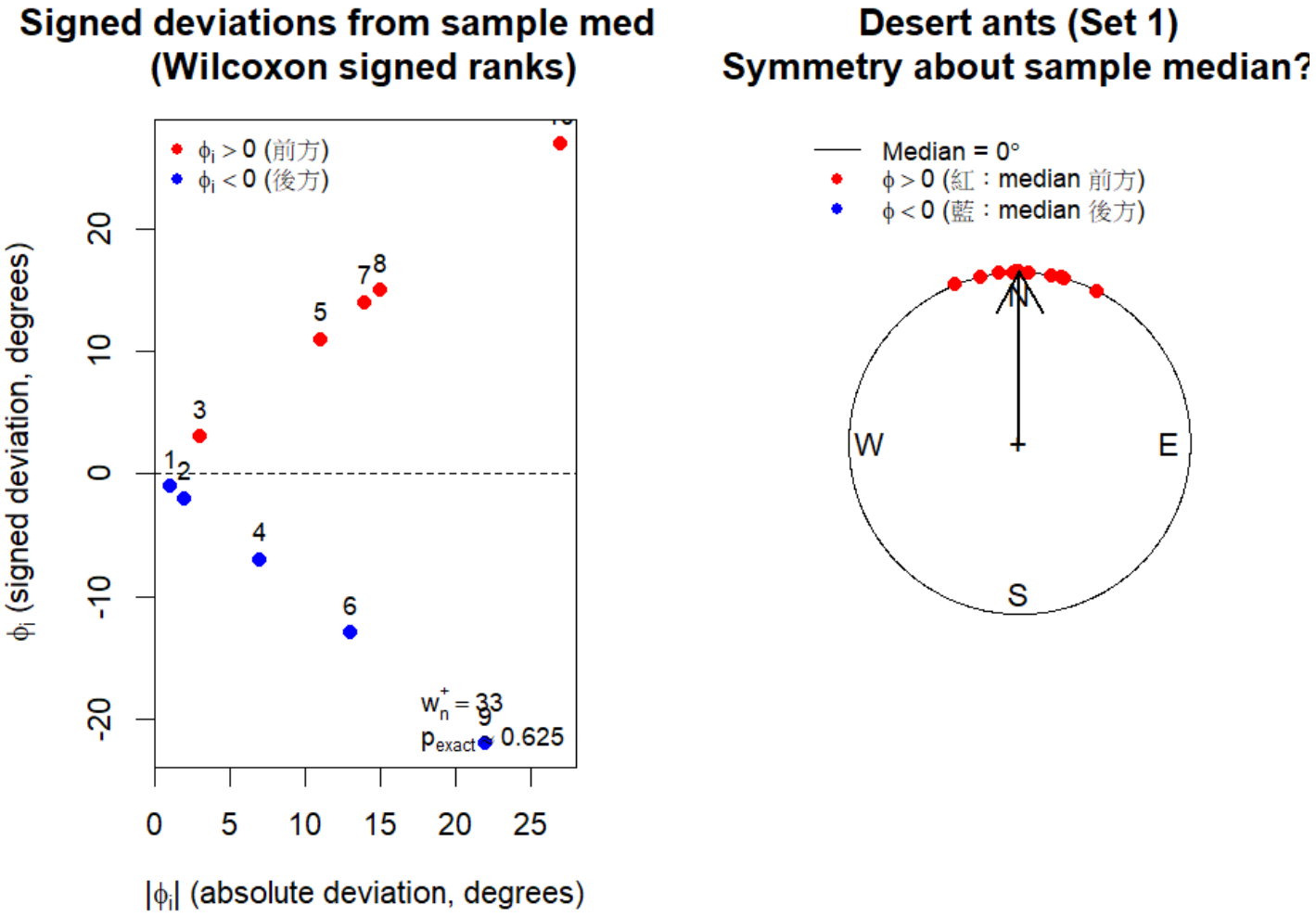


Figure: Exact Wilcoxon p-value (two-sided) = 0.625

Summary

- For small samples, prefer the median direction; use the arc–count confidence interval and a half–circle count test with chi–square reference.
- For moderate and large samples, use the mean direction; estimate a circular standard error from dispersion, then apply a sine–linearized confidence interval and a standard normal test.
- To assess symmetry about an unknown center, use the Wilcoxon signed–rank test on signed angular deviations.

Thank You!

References

Fisher, N. I. (1993). *Statistical Analysis of Circular Data*. Cambridge UP.

Proof of (4.19): CI width m for the median direction

Under the null that the true median is $\tilde{\mu}$, each point is equally likely to lie in either open semicircle centred at $\tilde{\mu}$. Ignoring ties,

$$M \sim \mathrm{Bin}(n, 1/2).$$

For a two-sided $100(1-\alpha)\%$ CI we require $\Pr\big(|M - n/2| \le c\big) \approx 1-\alpha$. By CLT,

$$\frac{M - n/2}{\sqrt{n/4}} \implies N(0,1) \quad \Rightarrow \quad c \approx \tfrac{1}{2} z_{1-\alpha/2}\sqrt{n}.$$

Counting c points on each side and converting to the book's "count-off" convention yields $m = 1 + \left\lfloor \frac{1}{2} z_{1-\alpha/2}\sqrt{n} \right\rfloor$, which is (4.19). Ties on the boundaries can be handled by reducing the effective n accordingly.

Proof of (4.20): Chi-square test for a specified median

With k observations exactly on the semicircle boundaries, the interior count $M \sim \mathrm{Bin}(n-k, 1/2)$ under $H_0 : \tilde{\mu} = \tilde{\mu}_0$. Then

$$\frac{M-(n-k)/2}{\sqrt{(n-k)/4}} \implies N(0,1). \quad \text{Thus} \quad \frac{\big(M-(n-k)/2\big)^2}{(n-k)/4} \implies \chi^2_1.$$

Writing $m = M$ and expanding the numerator gives

$$Y = \frac{\big(2m-(n-k)\big)^2}{n-k} = \frac{(2m-n+k)^2}{n-k} \implies \chi^2_1,$$

which is (4.20).

proof of (4.22): sine–linearised normality

Define

$$\bar{C} = \frac{1}{n}\sum_{i=1}^{n}\cos(\theta_i - \mu), \qquad \bar{S} = \frac{1}{n}\sum_{i=1}^{n}\sin(\theta_i - \mu), \qquad \rho = \mathbb{E}\{\cos(\theta - \mu)\} > 0.$$

Bivariate CLT.

$$\sqrt{n}\left[\begin{pmatrix}\bar{C}\\ \bar{S}\end{pmatrix} - \begin{pmatrix}\rho\\ 0\end{pmatrix}\right] \;\Rightarrow\; N\big(0,\ \Sigma\big).$$

Delta method. Let $g(c,s) = \dfrac{s}{\sqrt{c^2+s^2}}$. Then $\sin(\hat{\mu}-\mu) = g(\bar{C},\bar{S})$, $g(\rho,0) = 0$, and $\nabla g(\rho,0) = (0,\rho^{-1})$. Hence

$$\sqrt{n}\,\sin(\hat{\mu}-\mu) \;\Rightarrow\; N\Big(0,\ \nabla g(\rho,0)^\top \Sigma\, \nabla g(\rho,0)\Big) = N\Big(0,\ \rho^{-2}\,\mathrm{Var}\{\sin(\theta-\mu)\}\Big).$$

Since $\mathrm{Var}\{\sin(\theta-\mu)\} = (1-\rho_2)/2$ with $\rho_2 = \mathbb{E}\{\cos(2(\theta-\mu))\}$,

$$\sqrt{n}\,\sin(\hat{\mu}-\mu) \;\Rightarrow\; N(0,\delta), \qquad \delta = \frac{1-\rho_2}{2\rho^2}.$$

Slutsky. With $\hat{\sigma}^2 = \hat{\delta}/n$, $\hat{\delta} \xrightarrow{p} \delta$. Therefore,

$$\frac{\sin(\hat{\mu}-\mu)}{\hat{\sigma}} \;\Rightarrow\; N(0,1).$$

The covariance matrix Σ (used in the CLT)

By definition,

$$\Sigma = \begin{pmatrix} \mathrm{Var}\big(\cos(\theta-\mu)\big) & \mathrm{Cov}\big(\cos(\theta-\mu), \sin(\theta-\mu)\big) \\ \mathrm{Cov}\big(\cos(\theta-\mu), \sin(\theta-\mu)\big) & \mathrm{Var}\big(\sin(\theta-\mu)\big) \end{pmatrix}.$$

Using trigonometric identities and $\rho = \mathbb{E}\{\cos(\theta-\mu)\}$,

$$\mathrm{Var}\big(\sin(\theta-\mu)\big) = \frac{1-\rho_2}{2}, \qquad \mathrm{Var}\big(\cos(\theta-\mu)\big) = \frac{1+\rho_2}{2} - \rho^2,$$

$$\mathrm{Cov}\big(\cos(\theta-\mu), \sin(\theta-\mu)\big) = \tfrac{1}{2}\,\mathbb{E}\big\{\sin\big(2(\theta-\mu)\big)\big\}.$$

Remark. If the distribution is symmetric about μ, then $\mathbb{E}\{\sin(2(\theta-\mu))\} = 0$ and the off-diagonal elements vanish, but the proof above does not require this symmetry.

Proof of (4.22): sine–linearised CI for μ

Let $z_{\alpha/2}$ denote the upper $100(\alpha/2)\%$ point of $N(0,1)$ (Appendix A1). A two–sided $(1-\alpha)$ confidence region for μ in the *sine* scale is therefore

$$\left|\sin(\bar{\theta}-\mu)\right| \;\leq\; z_{\alpha/2}\,\hat{\sigma}.$$

Provided $z_{\alpha/2}\hat{\sigma}$ is not too large, so we can invert this inequality to obtain

$$\mu \in \big(\,\bar{\theta} - \arcsin(z_{\alpha/2}\hat{\sigma}),\; \bar{\theta} + \arcsin(z_{\alpha/2}\hat{\sigma})\,\big) \pmod{2\pi},$$

which is exactly the sine–linearised confidence interval given in (4.22).

Proof of (4.23): large–sample Z–test for $\mu = \mu_0$

To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we apply the same linearisation with the null value μ_0. Under H_0, the preceding arguments give

$$\frac{\sin(\bar{\theta} - \mu_0)}{\hat{\sigma}} \implies N(0, 1),$$

where $\hat{\sigma}^2 = \hat{\delta}/n$ is computed from the data as in (4.21).

Thus the test statistic

$$Z = \frac{\sin(\bar{\theta} - \mu_0)}{\hat{\sigma}}$$

is approximately standard normal under H_0. For a two–sided level-α test we therefore reject H_0 when

$$|Z| > z_{\alpha/2},$$

or, equivalently, when the corresponding two–sided p–value $2\{1 - \Phi(|Z|)\}$ is less than α. This is the large–sample Z–test stated in equation (4.23).

Proof of (4.24)–(4.27): Wilcoxon signed-rank test for symmetry

Let $\tilde{\mu}$ denote the sample median direction. Define centred angles

$$\phi_i \;=\; \theta_i - \tilde{\mu}, \qquad i = 1, \ldots, n,$$

Remove any $\phi_i = 0$, and let n' be the number of remaining observations.

For each nonzero ϕ_i, define the absolute-deviation ranks

$$r_i \;=\; \mathrm{rank}\big(|\phi_i|\big),$$

with average ranks for ties (as in the usual Wilcoxon signed-rank statistic). The sum of positive ranks is

$$w_{n'}^{+} \;=\; \sum_{\phi_i > 0} r_i,$$

Under the null hypothesis that the underlying distribution is symmetric about *some* direction, the signs $\mathrm{sgn}(\phi_i)$ are i.i.d. Rademacher (± 1 with probability $1/2$ each) and independent of the magnitudes $|\phi_i|$, hence independent of the ranks $\{r_i\}$. Therefore, conditional on the set of ranks, $w_{n'}^{+}$ has the *classical Wilcoxon signed-rank null distribution* with sample size n'.

Proof of (4.24)–(4.27): Wilcoxon signed-rank test for symmetry

Thus,

$$\mathbb{E}(w_{n'}^{+}) = \frac{n'(n'+1)}{4}, \qquad \mathrm{Var}(w_{n'}^{+}) = \frac{n'(n'+1)(2n'+1)}{24},$$

as stated below (4.27). For $n' < 16$, the exact critical values are tabulated (Appendix A7). For $n' \geq 16$, CLT tells,

$$Z = \frac{w_{n'}^{+} - \frac{n'(n'+1)}{4} + \frac{1}{2}}{\sqrt{\frac{n'(n'+1)(2n'+1)}{24}}} \implies N(0,1),$$

which yields the two-sided level-α rejection rule $|Z| > z_{\alpha/2}$.